

Project Design Phase Solution Architecture

Date	05 July 2026
Team ID	Solo Project – Individual Capstone
Project Name	Library Management System (ServiceNow)
Maximum Marks	4 Marks

Solution Architecture:

Smart Library Request Workflow Architecture: (built on ServiceNow PDI: dev218155)

The diagram below reflects the actual configuration verified on the personal Developer Instance: the Book (u_book) and Borrow Request (u_borrow_request) tables, the Borrow Request Approval Flow in Flow Designer, and the Access Control rules on both tables.

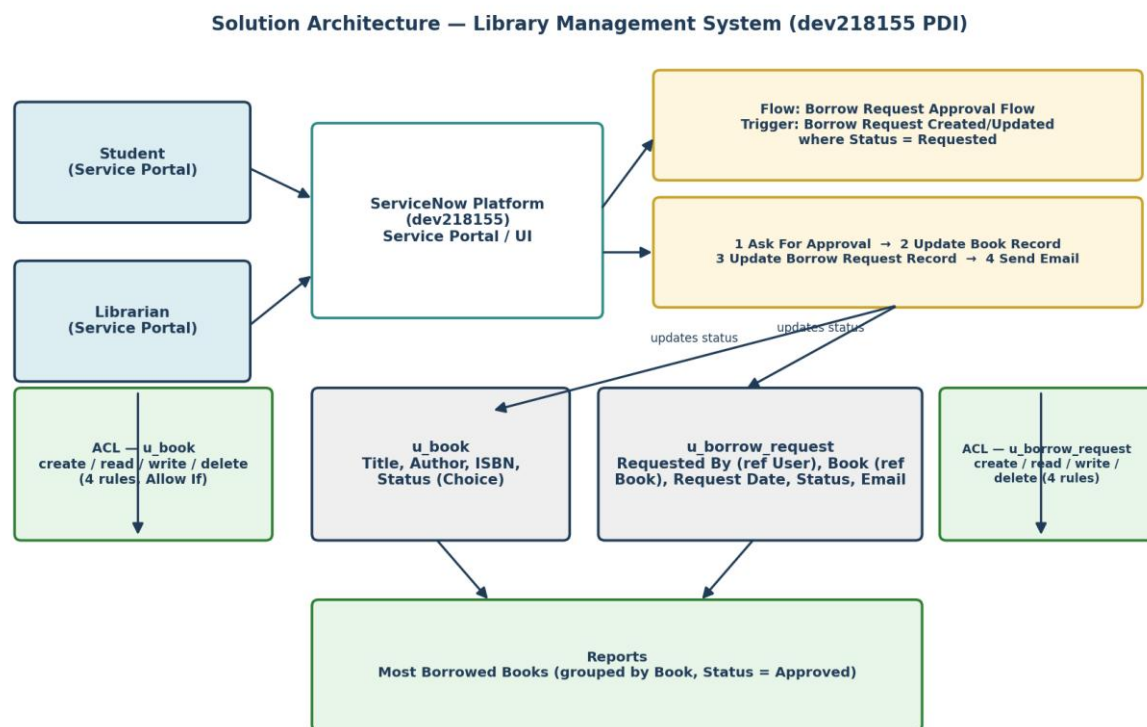


Table 1: Book (u_book) — Fields

Field	Type	Notes
Title	String	Book title
Author	String	Book author
ISBN	String	Book ISBN
Status	Choice	Available / Issued / Lost

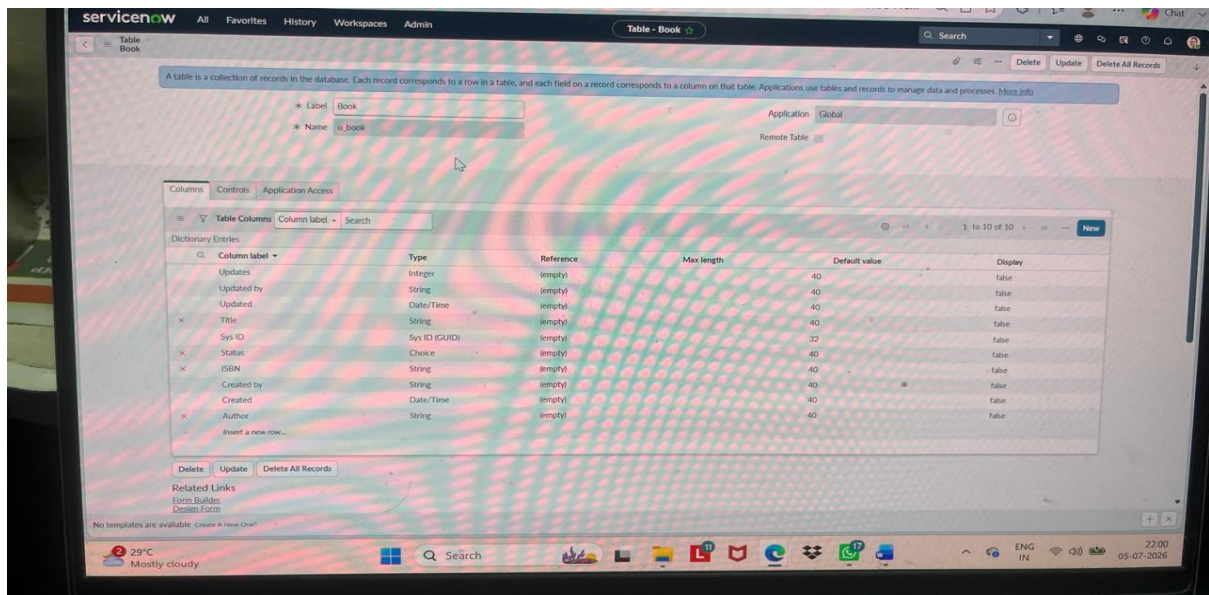


Table 2: Borrow Request (u_borrow_request) — Fields

Field	Type	Notes
Status	Choice	Requested / Approved / Rejected / Returned
Requested By	Reference (User)	Student who submitted the request
Email	String	Notification email captured for the request
Book	Reference (Book)	Book being requested
Request Date	Date/Time	Date/time the request was submitted

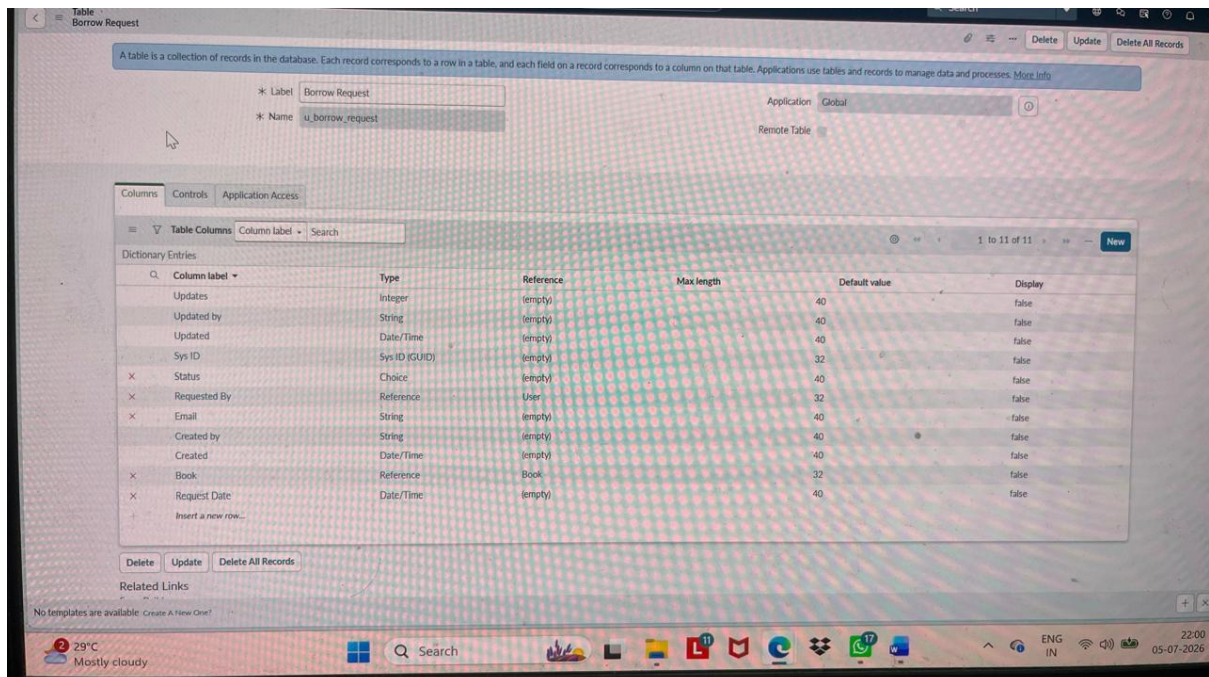
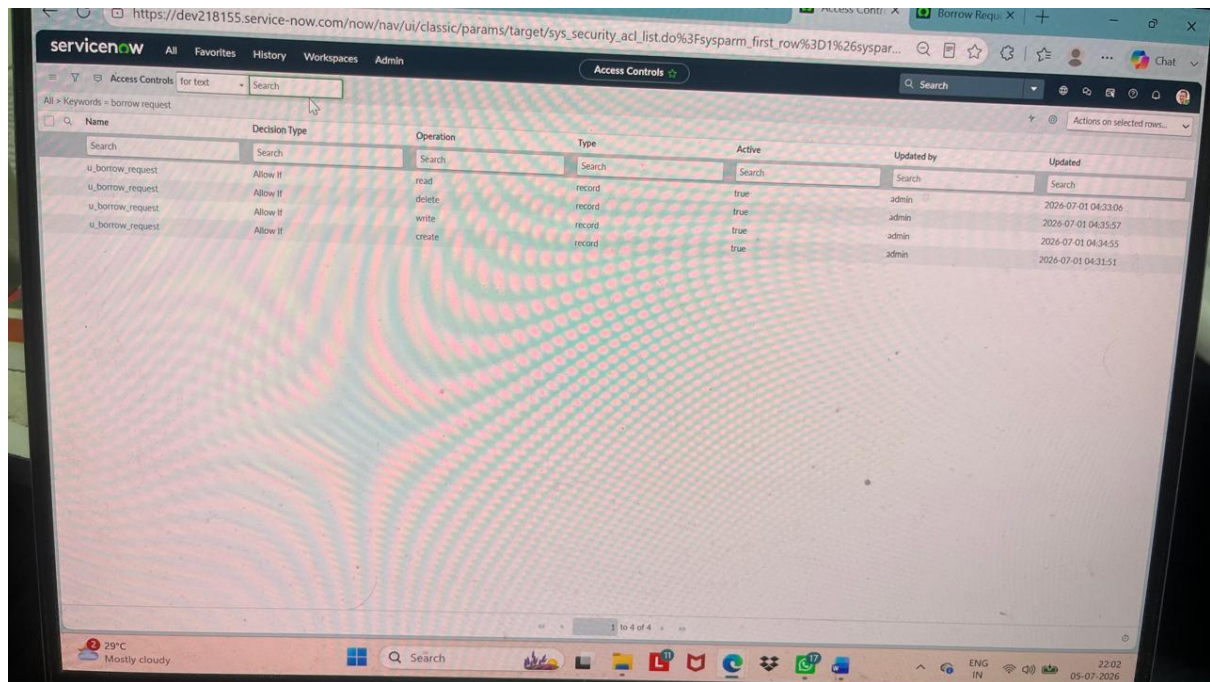


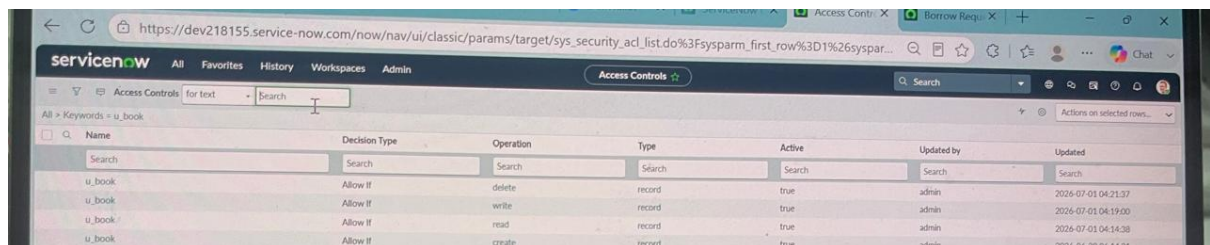
Table 3: Access Control (ACL) Summary

Table	Operation	Decision Type	Active
u_book	create	Allow If	true
u_book	read	Allow If	true
u_book	write	Allow If	true
u_book	delete	Allow If	true
u_borrow_request	create	Allow If	true
u_borrow_request	read	Allow If	true
u_borrow_request	write	Allow If	true
u_borrow_request	delete	Allow If	true

All 8 ACL rules were verified as Active (true) on both tables, each set to Allow If / record type.



Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_borrow_request	Allow If	read	record	true	admin	2026-07-01 04:33:06
u_borrow_request	Allow If	delete	record	true	admin	2026-07-01 04:35:57
u_borrow_request	Allow If	write	record	true	admin	2026-07-01 04:34:55
u_borrow_request	Allow If	create	record	true	admin	2026-07-01 04:31:51



Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_book	Allow If	delete	record	true	admin	2026-07-01 04:21:37
u_book	Allow If	write	record	true	admin	2026-07-01 04:19:00
u_book	Allow If	read	record	true	admin	2026-07-01 04:14:38
u_book	Allow If	create	record	true	admin	2026-06-29 06:54:31

Borrow Request Approval Flow (Flow Designer)

Flow name: Borrow Request Approval Flow — Status: Active

Trigger: Borrow Request Created or Updated, where Status is Requested.

1. Ask For Approval on Borrow Request
2. Update Book Record
3. Update Borrow Request Record
4. Send Email

An Error Handler is present on the flow (currently inactive) to catch failures in any of the four actions above.

